Elvira Kadaub

AI PRODUCT MANAGER · PRODUCT DATA SCIENTIST · FOUNDER

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SUMMARY

AI Product Manager and Product Data Scientist with 10+ years of experience building ML, data, and workflow automation products for complex domains. Former Genentech Senior Data Scientist / Product Manager with experience leading validated AI/data products across clinical, quality, manufacturing, and regulatory workflows. Founder & AI Product Lead at Insurain, building an AI-enabled insurance marketplace and agentic workflows for policy intake, quote routing, and customer follow-up.

EXPERIENCE

Founder & AI Product Lead · Insurain

Feb 2025 – Present

- Building an AI-enabled insurance marketplace that helps customers compare home, auto, and renters insurance quotes privately, without spam or pressure.
- Own product strategy, customer discovery, agent workflows, quote-request funnel, marketplace design, and GTM execution.
- Designed AI-assisted policy intake and parsing workflows to extract structured coverage data from uploaded insurance documents.
- Built agent-side workflows for quote assignment, turnaround tracking, carrier matching, quote submission, and customer selection.
- Grew early marketplace activity to ~220 monthly quote requests and 20 active agents.
- Reduced time-to-first-quote from 24–72 hours to under 24 hours through workflow design, matching logic, and agent process improvements.

Senior Data Scientist / Product Manager · Genentech

Sep 2022 – Feb 2025

South San Francisco, CA

- Led AI and data product development across 10+ molecule teams, partnering with scientists, clinical experts, quality, manufacturing, and regulatory stakeholders to modernize regulated drug-development workflows.
- Served as Product Owner for 5+ internal AI/data applications across quality control, manufacturing, clinical, and regulatory use cases — owning roadmap, prioritization, stakeholder communication, validation, and adoption.
- Led a cross-functional team of 5 engineers from MVP to validated production release, delivering Roche-adopted applications for complex scientific and operational workflows.
- Built validated, audit-ready data products supporting 5+ health-authority filings across 10+ molecules, improving traceability and reducing manual evidence-generation effort.
- Developed AutoML algorithms using Raman spectroscopy, chromatography, LC/MS, and process metadata to predict sample properties — reducing analysis workflows from days to minutes and improving speed by >99%.
- Automated LC/MS performance monitoring across 10 instruments and 2 labs, saving 40+ hours/week and accelerating failure detection, alerting, and root-cause analysis.
- Designed production ML evaluation frameworks using precision/recall, error analysis, domain-expert validation, and human-in-the-loop review to improve reliability in regulated environments.

Data Scientist, ADQC Innovation Pod · Genentech

Jun 2019 – Sep 2022

South San Francisco, CA

- Developed ML and automation solutions for analytical development and quality control workflows across biologics development.
- Built AutoML and feature-selection methods for Raman spectroscopy models, reducing manual analysis from days to minutes and enabling faster product quality assessment.
- Developed automated chromatography and LC/MS data-processing workflows to monitor system suitability, assay performance, and instrument health across regulated lab environments.

- Created dashboards, alerts, and data pipelines to improve visibility into analytical performance, failure patterns, and root-cause investigation.
- Translated scientific requirements into production-ready analytics tools, creating the foundation for broader Roche/Genentech workflow automation.

Data Science Research Specialist, Mathematical Oncology Lab · UC Irvine

Aug 2017 – Jun 2018

- Developed mathematical and computational models of cancer drug resistance.
- Built dynamical-systems and agent-based models to study melanoma treatment response and optimize dosing strategies.
- Collaborated with researchers on translational oncology modeling and scientific publication work.

Automation Software Engineer · Netcracker Technology

Oct 2013 – May 2015

- Built automation tools and software solutions supporting enterprise telecom workflows.
- Worked across scripting, testing, process automation, and technical delivery.

SELECTED PROJECTS

PolicyFlow Agent — Agentic AI workflow for insurance policy intake and quote prep: extraction, schema validation, missing-field detection, recommendations, and human-approved follow-up drafts. Python · FastAPI · LangGraph · Pydantic.

Insurain — AI-enabled marketplace for private home, auto, and renters quote comparison; AI policy parsing and agent workflows.

Clinical–CMC Data Product (Genentech) — Validated, audit-ready data product linking clinical exposure and CMC/quality data; supported 5+ health-authority filings.

AutoML for Analytical Quality (Genentech) — ML and automation over Raman, LC/MS, and chromatography; >99% faster analysis, 40+ hours/week saved.

SKILLS

PRODUCT	AI Product Management, Product Strategy, Product Ownership, Roadmapping, Stakeholder Management, User Discovery, Marketplace Design, Workflow Design, Metrics, GTM
AGENTIC AI / LLM	Agentic AI Workflows, LLM Applications, Structured Outputs, Tool Use, Human-in-the-loop Review, Evaluation, Prompting, RAG Concepts, LangGraph, Policy Parsing
DATA / ML	Python, SQL, pandas, scikit-learn, AutoML, ML Evaluation, Precision/Recall, Error Analysis, A/B Testing, Statistical Modeling, Dashboards, Data Pipelines
CLOUD / TOOLS	GCP, BigQuery, Tableau, FastAPI, Docker, Git, Streamlit, Next.js, Postgres, Snowflake
DOMAINS	Biotech, Regulated AI, Clinical–CMC Workflows, Quality Control, Manufacturing, Insurtech, Insurance Quote Workflows

EDUCATION

M.S. Applied Mathematics
University of Texas at Dallas

B.S. / M.S. Mathematics & Computer Science
Samara National Research University

CERTIFICATIONS

Machine Learning
Stanford

Scrum Master
Scrum Inc.

PUBLICATIONS & PATENTS

- PATENT **Use of Genetic Algorithms to Determine Sample Properties from Raman Spectra.** US20230009725A1.
- PATENT **Diagnostic Method for Determining the Presence of Label-Free Antibody–Antigen Complexes in Biological Samples.**
- PAPER **Multi-Attribute Raman Spectroscopy (MARS) for Monitoring Product Quality Attributes in Formulated Monoclonal Antibody Therapeutics.** mAbs, 2022; 14(1):2007564.